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Class-imbalanced dynamic financial distress prediction based on Adaboost-SVM ensemble combined with SMOTE and time weighting

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ABSTRACT

This paper focuses on how to effectively construct dynamic financial distress prediction models based on class-imbalanced data streams. Two class-imbalanced dynamic financial distress prediction approaches are proposed based on the synthetic minority oversampling technique (SMOTE) combined with the Adaboost support vector machine ensemble integrated with time weighting (ADASVM-TW). One is the simple integration model of SMOTE with ADASVM-TW, which uses SMOTE to make each data batch class-balanced before ADASVM-TW is applied for dynamic financial distress prediction modeling. The other is the embedding integration model of SMOTE with ADASVM-TW, which embeds SMOTE into the iteration of ADASVM-TW and creatively designs a new sample weighting mechanism. Namely, in each round of iteration, it does not only resample the majority financial normal samples by time weighting, but also generates more synthetic minority samples around new and difficult minority samples and less synthetic minority samples around old and easy minority samples to make the training dataset class-balanced. The empirical experiments are carried out based on the financial data of totally 2628 Chinese listed companies. Since there is certain degree of randomness in the proposed models, 50 times of experiments were performed under the same computational environment, so that the experimental results of model performance can be statistically compared and tested. The results indicate that both the simple integration model and the embedding integration model can greatly improve the recognition ability for the minority financial distress samples, and the embedding integration model is even more preferred because it also significantly outperforms the simple integration model.

1. Introduction

Financial distress means that when an enterprise suffers threat of external economic environment or internal financial decision failure, it will get into trouble due to insufficient cash flow, or even worse, it may suffer decline in profitability or even go bankrupt. During the period of the global financial crisis that erupted in 2008, many enterprises suffered serious losses and even became bankrupt. For an enterprise that lacks financial risk early warning mechanism, it is usually unable to treat the financial challenges and shocks properly when the outside economic environment deteriorates. To some extent, an enterprise's well-developed internal control mechanism should include a scientific financial distress prediction (FDP) system, which can signify financial deterioration before financial distress really comes. Therefore, FDP has been

widely studied and various approaches have been proposed to support more scientific financial risk management.

According to whether the FDP model is dynamically updated with time going on, FDP research is divided into two categories: i.e. stationary FDP modeling and dynamic FDP (DFDP) modeling. Most traditional FDP studies pay attention to stationary FDP modeling, which starts from the statistical analysis models such as Beaver's univariate model [1], Altman's Z-score model [2] and Ohlson's Logit model [3], and evolves to the artificial intelligence models such as decision tree [4], neural networks (NNs) [5], support vector machine (SVM) [6], case-based reasoning [7], etc. However, these early FDP models belong to single classifier models that may show instability in certain cases, and more recent studies on stationary FDP modeling focus on constructing multiple classifiers ensemble models by bagging, boosting, random forest, etc. [8–10]. These studies mainly focus on how to construct different kinds of

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classifiers for FDP based on a certain data set under stationary circumstance. The main conclusion is that artificial intelligence models usually show better FDP performance than statistical ones, and multiple classifiers ensemble can further improve the FDP performance by integrating the complementary information output by multiple base classifiers. But they neglect the phenomenon of financial distress concept drift and do not consider how to update FDP models dynamically. DFDP modeling based on data stream considers the effect of financial distress concept drift, and it is one of the hottest topics in the field of FDP in the recent decade. Some DFDP models have been proposed based on instance selection [11], batch-based adaptive dynamic classifier ensemble [12], incremental bagging classifier ensemble [13], time-weighting-based Adaboost classifier ensemble [14], etc. The empirical studies indicate the existence of financial distress concept drift and the superiority of DFDP models to stationary FDP models. The main reason is that DFDP models consider the change of data distribution over time, and attempt to treat it by dynamical model updating mechanisms.

In addition, the problem of class imbalance should be considered in FDP research, because financial distress enterprises are greatly less than financial normal enterprises in the real world. However, the above studies on DFDP modeling are all based on class-balanced datasets and neglect the effect of class imbalance. Some studies have paid attention to the class imbalance problem in case of stationary FDP modeling and find that the performance of most classification algorithms drops greatly when the datasets for FDP modeling are class-imbalanced. Particularly, the prediction accuracy for financial distress samples becomes especially low and the FDP models bias towards financial normal samples, which is more severe for artificial intelligence FDP models than statistical FDP models [15]. The most popular approaches for treating class imbalance in the literatures on stationary FDP modeling include the data-level methods such as over-sampling, under-sampling and synthetic minority over sampling technique (SMOTE), and the multiple classifier ensemble methods such as data-set-partition-based ensemble, SMOTE-Bagging [16], gradient boosting, random forests [17], deep-belief-network-based resampling ensemble [18], and heterogeneous ensemble [19]. They greatly improve the FDP performance for the minority financial distress samples under the circumstance of class imbalance, but they almost ignore how to integrate the class imbalance processing with time weighting in the process of DFDP modeling.

On one hand, the financial distress enterprises are much fewer than the financial normal enterprises in the real economic society, which evidently shows the characteristic of class imbalance. If a financial normal enterprise is misclassified as a financial distress enterprise in FDP, the misclassification cost is nothing more than the expense for the enterprise to strengthen its internal control. However, if a financial distress enterprise is recognized as a financial normal enterprise, it will miss the best time to take steps and may suffer serious losses or even go bankrupt. Therefore, it is of great significance to focus on class imbalance problem in the field of FDP and improve the recognition accuracy of financial distress samples. On the other hand, with the passage of time and the changing of economic environment, financial distress concept drift problem will be highlighted accordingly. Namely, with time going on, the definition of financial distress may change with the change of external economic environment and the evolving of enterprise life cycle, and at the same time, the inflow of new samples may lead to the change of data distribution. Such phenomena are defined as financial distress concept drift, which makes the FDP model built on the sample data of past period unsuitable for the current prediction. Therefore, it is necessary to constantly update the sample data with time going on and dynamically rebuild the FDP model with concept similar to the current financial distress.

However, most of the current FDP research is based on class balance or static modeling, although some studies are from the perspective of class imbalance, or try to deal with financial distress concept drift by dynamic FDP modeling. However, none of them consider the problem of class imbalance and dynamic FDP simultaneously. Hence, there is

an urgent need to propose an effective approach for class-imbalanced DFDP, which is the objective of this paper. This study attempts to consider concept drift and class imbalance simultaneously in the process of DFDP modeling. It is a further exploration based on Sun, Fujita, Chen, et al. (2017), which put forward an Adaboost SVM ensemble integrated with time weighting for class-balanced DFDP modeling with financial distress concept drift, abbreviated as ADASVM-TW [14]. It reconstructs a multi-SVM ensemble DFDP model at each new time point when a new data batch inflows, and effectively track the current concept of financial distress by a new sample weighting mechanism in Adaboost iteration, which integrates both thoughts of time-based weighting and error-based weighting. Considering the problem of class imbalance in DFDP, this paper tries to integrate SMOTE with ADASVM-TW in order to propose an effective class-imbalanced DFDP approach. Instead of simply applying the SMOTE algorithm to balance the initial training dataset batch by batch before the ADASVM-TW algorithm is run, which is named as simple integration of SMOTE with ADASVM-TW (S-SMOTE-ADASVM-TW), we embed SMOTE into the iteration process of ADASVM-TW, which is named as embedding integration of SMOTE with ADASVM-TW (E-SMOTE-ADASVM-TW). In the iteration process of E-SMOTE-ADASVM-TW, an improved sample weighting mechanism for coping with concept drift is used to guide the SMOTE processing of minority class samples and the selection of majority class samples, so as to construct multiple class-balanced training data sets for building base SVM classifiers, which can adapt to current financial distress concept. Therefore, this study will enrich the approaches for treating class-imbalanced data stream learning as well as the theoretical system of FDP approaches, and provide an important tool to aid practical financial risk management for enterprises.

This paper is organized as follows. Section 2 summarizes the current related literatures on FDP, class-imbalanced classification and concept drift, and points out the contribution of this study. Section 3 briefly introduces the background models including ADASVM-TW and S-SMOTE-ADASVM-TW. Section 4 theoretically and innovatively constructs the class-imbalanced DFDP models based on embedding integration of SMOTE with ADASVM-TW. Section 5 tests the proposed approaches by the empirical experiments and discusses the experimental results. Finally, conclusions are made in Section 6.

2. Literature review

Since research on class-imbalanced DFDP should consider the problem of class imbalance and concept drift simultaneously in the context of FDP, literature review is made respectively on FDP, class-imbalanced classification, and concept drift.

2.1. Literature review on FDP

FDP means predicting enterprises' financial distress by counting the sensitive key indicators or building statistical or artificial intelligence models. By now, different scholars have proposed various FDP models, which have improved the efficiency of FDP evidently.

In the early stage of FDP research, Beaver [1] proposes the single variable discriminant model, Altman [2] puts forward the multivariate discriminant analysis (MDA) model, and Ohlson [3] applies the logistic regression (Logit) model. These are the representative statistical FDP models and they are still widely applied today. Statistical models have the advantages such as simpler structure, less parameters and less training time, but they are restricted by statistical assumptions such as normal distribution and equal covariance matrixes of independent variables (e.g. MDA), and no multi-collinearity among independent variables (e.g. Logit). Then, with the development of artificial intelligence methods, Frydman et al. [4] construct a decision tree FDP model, and Odom and Sharda [5] construct an NNs FDP model, and they are early attempts of applying artificial intelligence methods to FDP. Varetto [20] further attempts to use genetic algorithm for FDP, but its performance is even

worse than that of MDA. Sun and Hui [7] propose a similarity weighted voting case-based reasoning model for FDP, and achieve better performance than MDA. In addition, SVM has been widely used for FDP after it is proposed by Vapnik [21]. Ding et al. [6] attempt applications of SVM to FDP and provide robust prediction accuracy for FDP, because SVM can effectively avoid the over-fitting problem due to small sample size. The artificial intelligence FDP models usually outperform statistical FDP models, because they are not restricted by statistical assumptions and can fit the non-linear relationship between financial distress and financial features better. Besides, decision tree and case-based reasoning FDP models are easier to be understood by corporate management, and NNs and SVM have stronger fitting ability although they are black-box models.

In recent years, hybrid classifier and multiple classifiers ensemble algorithm become the trend of developing FDP models. Mselmi et al. [22] integrate partial least squares (PLS) with SVM to propose the PLS-SVM FDP model, which makes use of PLS's ability for treating multicollinearity among predictors and SVM's ability for capturing nonlinear relationship of latent variables, and show that it achieves higher prediction accuracy than the Logit, NNs, SVM, and PLS when the sample data are from two years before financial distress. Barboza et al. [23] use various machine learning models such as SVM, bagging, boosting and random forest to predict financial distress one year prior to the event. Their results show that the accuracies of machine learning models are on average approximately 10% higher than those of traditional models, and classifier ensemble FDP models based on bagging, boosting, and random forest outperform the other techniques. Zhou et al. [24] take into consideration the four different listing statuses of Chinese listed companies, i.e. normal status without any risk warning, abnormal status with delisting risk warning, abnormal status with other risk warning, and delisted status, and propose a classifier ensemble architecture based on one-versus-one multi-class classification fusion using optimizing decision directed acyclic graph for predicting the companies' listing status. FDP models based on hybrid classifier and multiple classifiers ensemble have showed the superiority of making full use of complementary advantages of different algorithms or diverse base classifiers, and gradually turn to be the research focus of FDP modeling currently.

Besides, some scholars start their research on FDP from the industry point of view. For example, Holmes et al. [25] develop a potential indicator of financial distress including factors of unprofitability, equity decline, insolvency and close, and construct a model to predict the financial distress potential index for rural hospitals. Sayari and Mugan [26] use factor analysis and the concept of entropy to find the ratio set that best reflects the uncertainty level of a particular industry, and use the Logit model to build industry-specific FDP models. Li et al. [27] propose the mixed sample modeling approach that addresses the ranking information within samples in deficient information environments, and improves forecasting accuracy of predicting hospitality firm failure. Li et al. [28] put forward a new case-based deep-layer predictive analysis methodology for failure analysis of corporations with multiple hospitality businesses. Industry-oriented FDP models can capture the features of financial distress for a specific industry better, and therefore output more accurate FDP results for the enterprises of the industry. However, the FDP sample data of some industries are limited, which may restrict the FDP research of such industries.

2.2. Literature review on class-imbalanced classification

Class imbalance means that the sample numbers of different classes in the dataset are evidently unequal. When there are only two classes, we call the larger number of samples as the majority class and the smaller number of samples as the minority class. In fact, in many cases we are more concerned with the minority class, because the cost of misclassifying the minority class is usually much higher than the majority class [30]. Louzada et al. [31] use Brazil bank data to conduct an empirical study and find that the prediction results of the Logit models are signif-

icantly different based on different ratios of default samples and non-default samples. Therefore, class imbalance has evident impact on FDP performance and class-imbalance-oriented FDP approaches are worth exploring. Fernández et al. offers a comprehensive review of imbalanced learning and categorize it into data level approaches, algorithm level approaches, ensemble learning approaches, etc. [29].

As for the problem of class imbalance, SMOTE is one of the most influential data preprocessing technique. Chawla et al. [33] propose the SMOTE algorithm, which generates new samples by finding k nearest neighbors for every minority class sample and randomly interpolating between them, so as to achieve sample class balance before training classifiers. Fernandez et al. consider SMOTE as "de facto" standard in the framework of learning from imbalanced data due to its simplicity and robustness [32]. Li and Sun [34] propose an oversampling method to generate minority class samples based on nearest-neighbor support vectors for SVM modeling, and its application to the Chinese tourism companies' FDP obtains a high testing accuracy. In addition to the oversampling method, the under-sampling method can also be used to eliminate partial majority class samples. Tang et al. [35] improve the under-sampling method and propose a novel granular SVMs-repetitive under-sampling algorithm (GSVM-RU) that combines SVM with the granular technology. It uses SVM to generate support vectors, and the support vectors of majority class samples are under-sampled and combined with the minority class samples to form a balanced training data set. The experimental results show that the method performs well. Li et al. [36] propose the rebalanced and clustered SVM approach for restructuring performance prediction with skewed sample distribution, and the empirical results demonstrate its applicability for forecasting effectiveness of asset restructuring. Cordon et al. [37] introduce a newly developed novel package written in R and C++, for preprocessing of imbalanced dataset with some relatively new oversampling algorithms such as majority weighted minority oversampling [38], rapidly converging gibbs (RACOG) and wrapper-based RACOG [39], random walk oversampling [40], and probability distribution density function estimation based oversampling [41]. Data level approaches for coping with class imbalance are applicable to various classification algorithms, and can be widely used in all kinds of domains of class-imbalanced classification or prediction including FDP.

Besides the above class balancing approaches at the data level, there are also some approaches at the algorithm level and have achieved some success. Cleofas-Sánchez et al. [42] adopt the method called hybrid associative memory with translation based on NNs for FDP, and it shows good performance for FDP under class imbalance and data overlapping conditions. Raghuvanshi and Shukla [43] propose a novel class-specific kernelized extreme learning machine (ELM) that addresses the class imbalance problem more effectively. The results show that this approach can achieve better performance than tradition ELM approaches for class-imbalanced real-world datasets and has considerably lower computational cost. Algorithm level approaches are proposed by changing the internal theoretical principles of a certain classification algorithm, which makes it more complex to be understood and applied. Hence, research on algorithm level approaches for class-imbalanced FDP is relatively limited.

In addition, the classifier ensemble algorithm can also improve the classification performance of imbalanced dataset. Brown and Mues [17] empirically compare the performance of gradient boosting, least square SVM (LS-SVM), random forests, Logit, NNs and decision tree for loan default prediction under different degrees of class imbalance. The results indicate that random forest and gradient boosting classifiers perform better, and the classifier ensemble approaches are more robust than single classifiers for class imbalanced data sets. Chawla et al. [44] combine the Boosting algorithm with SMOTE method without changing the samples' weighting and resampling process of the Boosting iteration. Instead, the data balance is achieved by using SMOTE method for the selected samples and the results show good classification performance. Sun et al. [16] propose a SMOTE-Bagging-SVM ensemble method for

class-imbalanced FDP, which is proved to outperform SMOTE-SVM and data-set-partition-based ensemble. Sun et al. [48] put forward a decision tree ensemble based on SMOTE and bagging with differentiated sampling rates, and the empirical experiment proves its effectiveness for class-imbalanced FDP. Hence, integrating SMOTE with traditional classifier ensemble algorithms is one of the ways to further improve the ensemble learning approaches for class-imbalanced classification. González et al. [45] introduce a novel ensemble approach based on Switching with a new technique to select the switched examples based on Nearest Enemy Distance, which can balance the class distribution, reduce the induced noise, and maintain the minority class information, to result very promising performance. Recently, some literatures begin to focus on ensemble classifiers for the class imbalance problems in the scenario of multi-class classification. García et al. [46] develop a novel and effective ensemble classifier approach called as DES-MI (dynamic ensemble selection for multi-class imbalanced datasets), which consists of two key components: the generation of a balanced training dataset and the selection of appropriate classifiers. Zhang et al. [47] propose a novel approach named as DRCW-ASEG (Distance-based Relative Competence Weighting with Adaptive Synthetic Example Generation) for multi-class imbalanced classification problem, which properly addresses the synergy between imbalance learning and dynamic classifier weighting in one-versus-one scheme to construct ensemble classifiers. Therefore, classifier ensemble approaches for class-imbalanced multi-class classification is attracting more and more research attention, and it has become an important future research topic in the field of class-imbalanced classification.

2.3. Literature review on concept drift

Schlimmer and Granger (1986) [49] first propose the theory of concept drift. Concept drift is the phenomenon that target concept or the distribution of dataset changes as the data stream flows in over time [50]. Concept drift exists in many practical domains, and some studies have empirically test the existence of financial distress concept drift [11–14]. As for disposing concept drift, there are mainly three methods, i.e. the sample selection method, the time weighting method and the classifier ensemble method. Mitchell et al. [51] present the fixed-width time window method, which adopts the fixed-size sliding window to select training samples, and it can solve the concept drift problem to some extent. Klinkenberg [52] proposes the adaptive time weighting method. It uses past data batches to test the model that is constructed based on the current data batch, and then weights each batch of data according to the testing accuracy. It can make full use of samples whose concept is similar to the current concept, and thus improve prediction accuracy. Sun and Li [11] propose a batch data selection method, which selects samples that have similar concept to the current data batch, to construct the FDP model based on SVM. The results show that the prediction accuracy of this method is higher than the full memory time window method.

In addition to the above methods, the classifier ensemble algorithms are very important approaches for dealing with concept drift [53]. Wu et al. [54] propose an adaptive ensemble classifier through the use of Bagging technology to build base classifiers and use feature selection method to filter out part of the base classifiers to construct the final ensemble classifier. The results show that the method achieves good classification performance. Sun et al. propose a data-batch-based adaptive and dynamic SVM ensemble with time window for FDP in case of financial distress concept drift [12]. Liu and Wu use incremental bagging and genetic algorithm to construct incremental bagging classifier ensemble for dynamic forecasting of financial distress with concept drift and empirically test it with Chinese listed corporations [13]. Sun et al. [14] propose the ADASVM-TW approach to dynamically construct the SVM ensemble classifier for FDP, and the empirical results show that the model can effectively deal with the concept drift problem of financial distress. Besides et al. [55] propose an adapting dynamic classifier selection for concept drift, which dynamically select the custom ensemble

for test instances by considering both region dependency and time dependency. The experimental results show that this approach performs well in most cases requiring almost no parameter tuning. Therefore, the classifier ensemble approaches cope with concept drift by selecting the base classifiers that are more adaptable to the current concept for final integration, or by adaptively selecting or weighting the training samples to train base classifier that better reflect the current concept.

According to the above literature review, classifier ensemble approaches are becoming the trend of FDP research and the important solutions for coping with both class imbalance and concept drift. However, most of the current FDP research is based on class balance or static modeling, and some studies only consider class imbalance or DFDP with financial distress concept drift separately. None of them consider the problem of class-imbalanced DFDP and integrate SMOTE and time weighting into the inner iteration of Adaboost-SVM ensemble for class-imbalanced classification with concept drift. Therefore, the contribution of this study is to propose an effective approach for class-imbalanced DFDP based on the embedding integration of SMOTE with ADASVM-TW, which also enriches the classifier ensemble approaches for non-stationary class-imbalanced classification with concept drift.

3. Background models

In this section, the basic ADASVM-TW model without the treating mechanism of class imbalance and the simple integration model of SMOTE with ADASVM-TW are respectively introduced.

3.1. Basic ADASVM-TW model for DFDP

The time weighting method proposed by Klinkenberg [52] can be widely used to deal with the problem of concept drift, and the basic idea is that the importance of data batches increases monotonically with time. The concrete time weighting function is shown in formula (1).

$$w_t = \exp(-\lambda t) (t = 0, 1, 2, \dots, n-1) \quad (1)$$

In this formula, t denotes the data batch number so that the current data batch number is 0, and from the new to the old, the number of data batches increase in turn. Besides, n is the total number of the dataset batches, and the range of λ is [0, 0.99].

In order to solve the problem of concept drift in FDP, Sun et al. [14] propose the ADASVM-TW model, which improves the sample weighting mechanism in the iteration process of training base classifiers for the Adaboost-SVM ensemble. With the initial average sample weights, the improved sample weighting functions are shown in formula (2)–(4).

$$w_1^i = \frac{1}{m} (i = 1, 2, \dots, m) \quad (2)$$

$$w_{u+1}^i = w_u^i \exp(-\alpha_u^i \exp(\lambda t^i l_u^i)) (i = 1, 2, \dots, m; u = 1, 2, \dots, U-1) \quad (3)$$

$$l_u^i = \begin{cases} 1 & \text{if } f_u(x^i) = y^i \\ -1 & \text{if } f_u(x^i) \neq y^i \end{cases} \quad (4)$$

In the above formulas, U is the total number of iterations in the ADASVM-TW algorithm and m denotes the total sample number. w_{u+1}^i and w_u^i mean the weights of the i th sample respectively for the $(u+1)$ th and u th iterations, and l_u^i is used to control the adjustment direction of sample weights. A value of 1 is assigned to the correctly classified sample, and a value of -1 is assigned to the misclassified sample, so that the weights of the correctly classified samples are decreased and the weights of the misclassified samples are increased. Besides, α_u is the weight factor determined by the u th base classifier's error rate e_u , which is also the voting weight of the u th base classifier for the Adaboost ensemble.

$$\alpha_u = 0.5 \ln \left[\frac{1 - e_u}{e_u} \right] \quad (5)$$

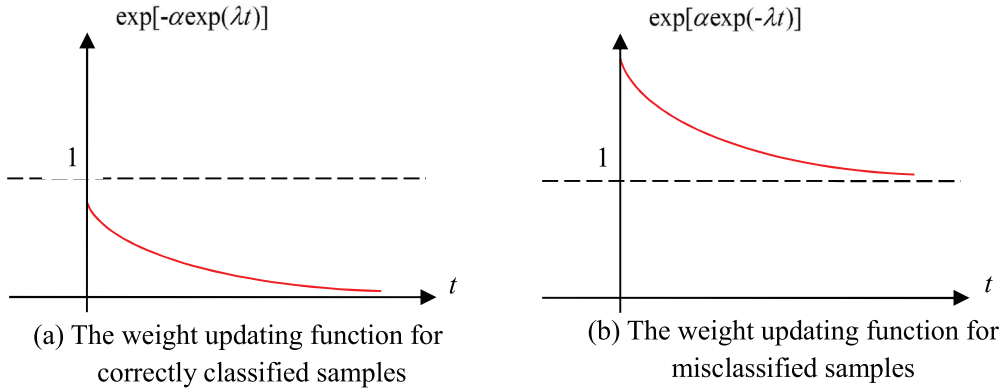


Fig. 1. The sample weight updating mechanism of ADASVM-TW.

Table 1

The algorithm of the basic ADASVM-TW model for DFDP.

Input: Initial training dataset $D = \{(x^i, y^i, t^i)\}, y^i \in \{1, -1\}, t^i \in \{0, 1, \dots, n-1\}, (i = 1, 2, \dots, m)$ and the error rate threshold c .

(1) Initialize the sample weights $w_1^i = \frac{1}{m}, W_1 = \{w_1^1, w_1^2, \dots, w_1^i, \dots, w_1^m\} = \{\frac{1}{m}, \frac{1}{m}, \dots, \frac{1}{m}\}$

(2) **For** $u = 1, 2, 3, \dots, U$

a) Select a training data subset D_u from D according to the sample weights W_u .

b) Train a base classifier SVM_u , also represented as $f_u(x)$, on the dataset D_u .

c) Validate SVM_u on the dataset D to calculate the classification error rate e_u .

d) If $e_u \leq c$ or $e_u \geq 0.5$, keep u unchanged and return to step a), otherwise continue to step e).

e) Calculate the voting weight of SVM_u : $\alpha_u = 0.5 \ln[\frac{1-e_u}{e_u}]$.

f) Update the sample weights: $w_{u+1}^i = w_u^i \exp(-\alpha_u t_u^i \exp(\lambda t_u^i)) (i = 1, 2, 3, \dots, m)$.

$$t_u^i = \begin{cases} 1 & \text{if } f_u(x^i) = y^i \\ -1 & \text{if } f_u(x^i) \neq y^i \end{cases}$$

g) Normalize the sample weights: $w_{u+1}^i = w_{u+1}^i / \sum_{j=1}^m w_{u+1}^j$

End

Output: ADASVM-TW ensemble $H(x) = \text{sign}(\sum_{u=1}^U (\alpha_u f_u(x)))$

The sample weight updating mechanism of ADASVM-TW is shown in Fig. 1. From the figure, we can intuitively find that, when a sample is correctly classified, its weight will be decreased, and the weights of newer samples will be decreased less than older ones. Meanwhile, when a sample is wrongly classified, its weight will be increased and the increase magnitude of newer samples is larger than older ones. Therefore, ADASVM-TW can provide more training opportunities for misclassified samples and newer samples, which helps to improve the DFDP performance. The algorithm of ADASVM-TW is shown in Table 1 and the detailed explanation can be found in Sun et al. [14].

The ADASVM-TW model updates the weights of training samples with the improved sample weight updating mechanism, which increases the weights of misclassified newer samples more than misclassified older samples, and reduces the weights of correctly classified newer samples less than correctly classified older samples. Therefore, it can empower higher weights to misclassified samples and newer samples and thus improve the model's adaptability to dynamic data flow. However, since the model does not consider the problem of class imbalance, it is not applicable to class-imbalanced dynamic data flow and thus cannot provide satisfactory support for class-imbalanced DFDP.

3.2. Simple integration of SMOTE with ADASVM-TW

As one of the commonly used oversampling methods for handling class imbalance, SMOTE artificially generates new minority class samples to balance the training data set, so as to improve the classification performance of the model. Namely, the new minority class samples generated by SMOTE are then combined with the original training set to obtain a balanced training dataset. As a simple idea of applying SMOTE to treating class imbalance for ADASVM-TW, the SMOTE algorithm can be applied firstly to balance the initial training dataset batch by batch before the ADASVM-TW algorithm is run, which is named as simple integration of SMOTE with ADASVM-TW (S-SMOTE-ADASVM-TW). For

the DFDP modeling, the initial training dataset is composed of a series of data batches that are collected at time intervals. Each data batch is labeled with the data batch number t with the same meaning as in formula (1). Just as the algorithm in Table 1 shows, the training samples in different data batches have different values of t , and take different roles in the process of ADASVM-TW model construction. Therefore, SMOTE is applied to each data batch respectively instead of the whole initial training dataset. The flow chart of S-SMOTE-ADASVM-TW is shown in Fig. 2.

As Fig. 2 shows, for the current modeling year, suppose there are n class-imbalanced data batches available for DFDP modeling, and they are represented as data batch $n-1, n-2, \dots, 1, 0$. For each data batch, SMOTE is carried out to generate synthetic financial distress samples, so that the class-balanced data batches $n-1, n-2, \dots, 1, 0$ are obtained and they are combined as the class-balanced training data set D^* . Then the basic ADASVM-TW algorithm illustrated as Table 1 is carried out on D^* to construct the DFDP model.

The S-SMOTE-ADASVM-TW model is expected to produce better performance than the basic ADASVM-TW model for class-imbalanced DFDP, because it introduces SMOTE for the data level treatment of skewed class distribution of the training dataset and makes it balanced before ADASVM-TW modeling, in which the base SVM classifier is sensitive to class imbalance problem. However, the process of SMOTE and the process of ADASVM-TW modeling are two relatively separate stages when they are simply integrated, and the process of SMOTE does not consider the different roles of different data batches for DFDP. Namely, the minority samples in older data batches and in newer data batches are expanded at the same degree without considering their ages. This is not consistent with the assumption of DFDP that new samples reflect more information of the current financial distress concept and are more important for the current FDP modeling. At the same time, since the process of SMOTE is outside the iteration process of ADASVM-TW, it does not treat the easy samples and the difficult samples differently when

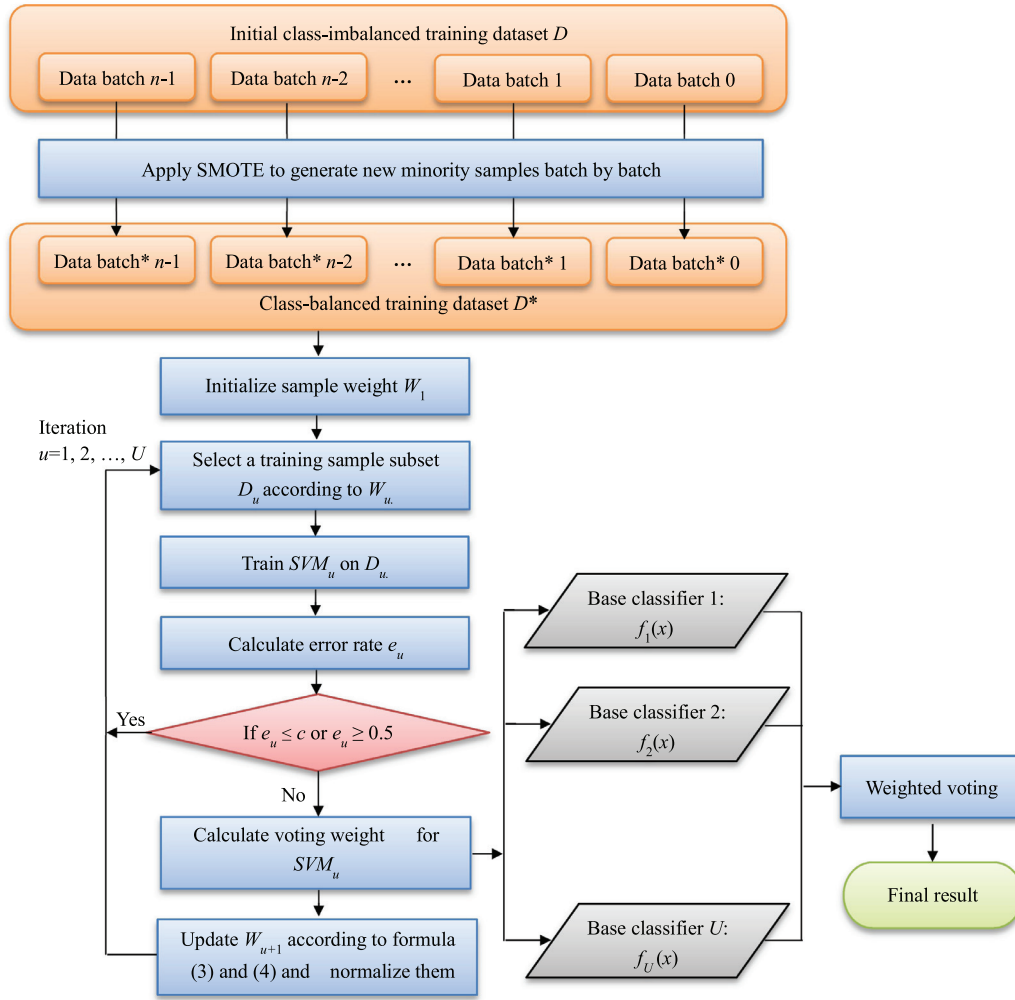


Fig. 2. The flow chart of S-SMOTE-ADASVM-TW model.

generating synthetic minority samples, which is also not consistent with the idea of Adaboost that difficult samples should be emphasized in the process of training a classifier ensemble.

4. Class-imbalanced DFDP models based on embedding integration of SMOTE with ADASVM-TW

For the purpose of training a good Adaboost SVM ensemble for class-imbalanced DFDP, it is more rational to embed SMOTE into the iteration process of ADASVM-TW, treating newer minority samples and older minority samples differently as well as treating the difficult and easy minority samples differently, which is named as embedding integration of SMOTE with ADASVM-TW (E-SMOTE-ADASVM-TW).

4.1. Framework of E-SMOTE-ADASVM-TW

Unlike the traditional SMOTE algorithm that generates the same number of synthetic minority samples around each original minority sample, the SMOTE process in E-SMOTE-ADASVM-TW tries to generate more synthetic minority samples around newer and difficult minority samples, and generate less synthetic minority samples around older and easy minority samples. Through this way, the classification knowledge hidden in newer and difficult minority samples is expanded with more degree and is better learnt to train base classifiers, which can well reflect the current financial distress concept as well as discriminate the difficult minority samples accurately. For this purpose, the SMOTE process is

embedded into the iteration process of ADASVM-TW, and the improved sample weighting mechanism of ADASVM-TW is adopted to determine the number of synthetic minority samples that should be generated by SMOTE around each original minority sample.

As formula (3) shows, the sample weights are updated by considering both the age and the difficulty level of different samples. As stated in Table 1, the improved sample weighting mechanism plays an important role in selecting training data subset in the iteration of ADASVM-TW algorithm based on a class-balanced dataset. However, in the iteration of E-SMOTE-ADASVM-TW algorithm for class-imbalanced DFDP, the improved weights of the majority samples and those of the minority samples have different functions. For the majority class, the improved weights are utilized to select the training subsets of majority samples. While, for the minority class, the improved weights are utilized to guide the generation of synthetic minority samples by SMOTE. Therefore, the majority samples' updated weights should be normalized among the majority class, and the minority samples' updated weights should be normalized among the minority class. The flow chart of E-SMOTE-ADASVM-TW is shown in Fig. 3.

4.2. Algorithm of E-SMOTE-ADASVM-TW

The specific algorithm of E-SMOTE-ADASVM-TW is shown in Table 2. Suppose the number of minority samples in the initial training dataset is denoted as S_p , and that of majority samples is denoted as S_N . Let $P_{-w_u^j}$ represent the normalized weight of the j th minority sam-

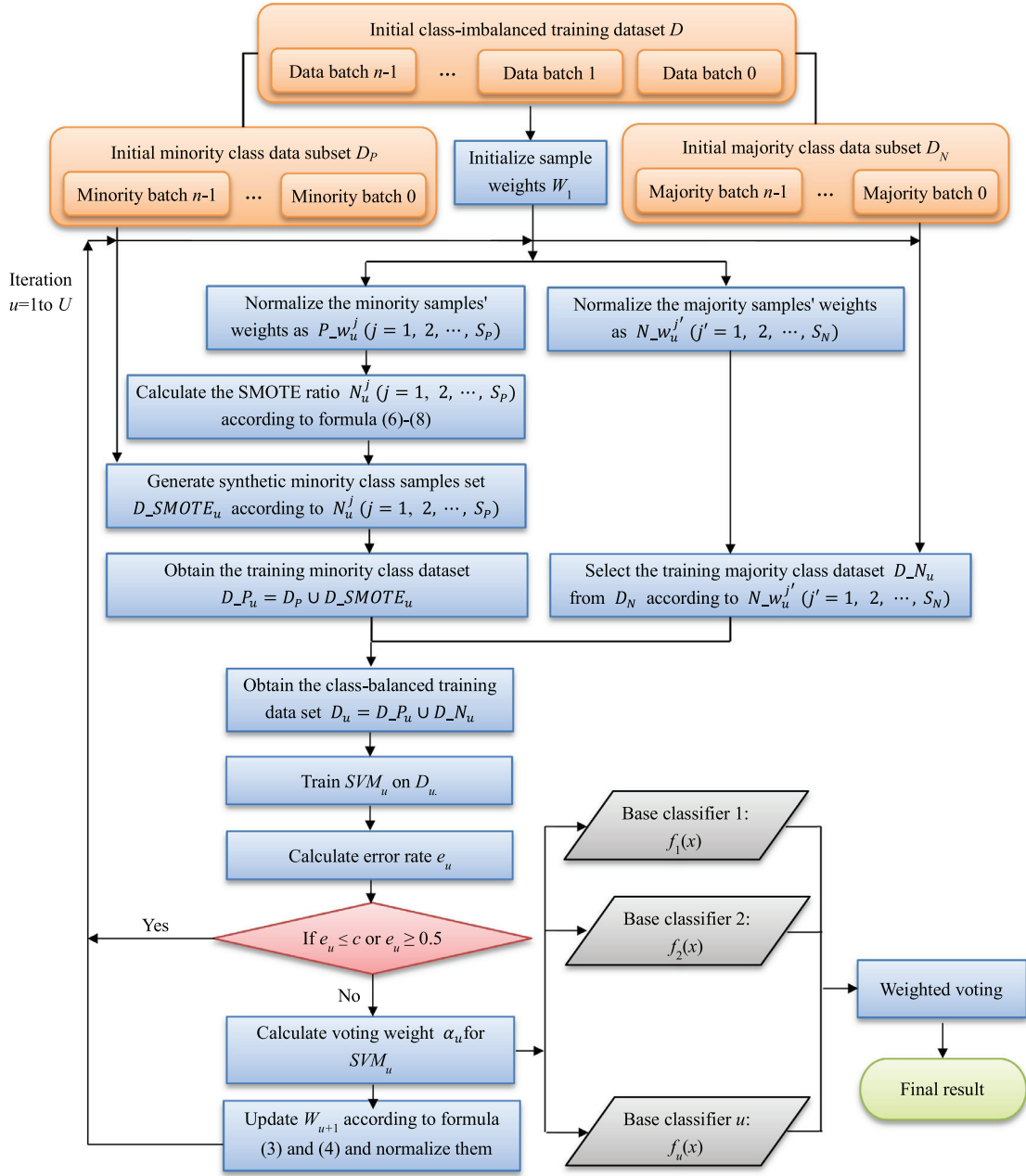


Fig. 3. The flow chart of E-SMOTE-ADASVM-TW model.

ple in the u th iteration of SVM ensemble training, and all the minority samples and their weights are sorted in terms of the descending order of $P_w^j (j = 1, 2, \dots, S_P)$. To make the training data set balanced, the total number of new minority class samples that need to be generated by SMOTE (denoted as S_{N-P}) equals to the majority sample number minus the minority sample number, i.e. $S_{N-P} = S_N - S_P$. Then the number of new samples that need to be generated around each minority class sample can be calculated according to formula (6)–(8).

$$N_u^j = \text{round}(S_{N-P} \times P_{w_u}^j) \quad (j = 1, 2, \dots, S_P) \quad (6)$$

$$M_u = \begin{cases} S_{N-P} - \sum_{j=1}^{S_P} N_u^j & \text{if } S_{N-P} > \sum_{j=1}^{S_P} N_u^j \\ \sum_{j=1}^{S_P} N_u^j - S_{N-P} & \text{if } S_{N-P} < \sum_{j=1}^{S_P} N_u^j \end{cases} \quad (7)$$

$$\begin{cases} N_u^j = N_u^j + 1 & \text{if } S_{N-P} > \sum_{j=1}^{S_P} N_u^j \\ (j = 1, 2, \dots, M_u) \\ N_u^j = N_u^j - 1 & \text{if } S_{N-P} < \sum_{j=1}^{S_P} N_u^j \\ (j = S_P - Z - M_u + 1, S_P - Z - M_u + 2, \dots, S_P - Z) \end{cases} \quad (8)$$

In formula (6), $\text{round}(\cdot)$ is the function for obtaining the round number, and N_u^j represents the number of new samples that need to be generated around the j th minority sample in the u th iteration of SVM ensemble training. By multiplying S_{N-P} by $P_{w_u}^j$, the total number of new synthetic minority class samples is allocated to each original minority class sample. Namely, the difficult and newer minority class samples with higher weights have priority to generate more synthetic minority class samples around them than the easy and older ones with lower weights. In most cases, $S_{N-P} \times P_{w_u}^j$ is a decimal number instead of an integer, and the function $\text{round}(\cdot)$ can help it round up

Table 2

The algorithm of E-SMOTE-ADASVM-TW.

Input: Initial training dataset $D = \{(x^i, y^i, t^i)\}, y^i \in \{1, -1\}, t^i \in \{0, 1, \dots, n-1\}, (i = 1, 2, \dots, m)$, which is divided into two subsets: the minority class subset $D_P = \{(x_P^j, y_P^j, t_P^j)\}, y_P^j = 1, t_P^j \in \{0, 1, \dots, n-1\}, (j = 1, 2, \dots, S_P)$ and the majority class subset $D_N = \{(x_N^{j'}, y_N^{j'}, t_N^{j'})\}, y_N^{j'} = -1, t_N^{j'} \in \{0, 1, \dots, n-1\}, (j' = 1, 2, \dots, S_N)$, namely $D = D_P \cup D_N, m = S_P + S_N$. The error rate threshold ϵ .	
Do:	
(1) Initialize the sample weights $w_1^i = \frac{1}{m}, W_1 = \{w_1^1, w_1^2, \dots, w_1^m\} = \{\frac{1}{m}, \frac{1}{m}, \dots, \frac{1}{m}\}$	
(2) For $u = 1, 2, 3, \dots, U$	
a) Respectively normalize the minority samples' weights as $P_{-w_u}^j (j = 1, 2, \dots, S_P)$ and majority samples' weights as $N_{-w_u}^{j'} (j' = 1, 2, \dots, S_N)$.	
b) Calculate the number of new synthetic samples that need to be generated around each minority class sample, namely the SMOTE ratios $N_u^j (j = 1, 2, \dots, S_P)$, according to formula (6)–(8).	
c) Generate N_u^j synthetic minority class samples around each original minority sample x_P^j by SMOTE and they constitute D_SMOTE_u .	
d) Obtain the training dataset of minority class $D_{-P_u} = D_P \cup D_SMOTE_u$.	
e) Select the training dataset of majority class D_{-N_u} from D_N according to the majority samples' weights $N_{-w_u}^{j'} (j' = 1, 2, \dots, S_N)$.	
f) Obtain the class-balanced training data set $D_u = D_{-P_u} \cup D_{-N_u}$.	
g) Train a base classifier SVM_u , also represented as $f_u(x)$, on the dataset D_u .	
h) Validate SVM_u on the dataset D to calculate the classification error rate e_u .	
i) If $e_u \leq \epsilon$ or $e_u \geq 0.5$, keep u unchanged and return to step a), otherwise continue to step j).	
j) Calculate the voting weight of SVM_u : $\alpha_u = 0.5 \ln[\frac{1-e_u}{e_u}]$	
k) Update the sample weight: $w_{u+1}^i = w_u^i \exp(-\alpha_u t^i \exp(\lambda t^i l_u^i)) (i = 1, 2, 3, \dots, m)$	
$l_u^i = \begin{cases} 1 & \text{if } f_u(x^i) = y^i \\ -1 & \text{if } f_u(x^i) \neq y^i \end{cases}$	
l) Normalize sample weights $w_{u+1}^i = w_{u+1}^i / \sum_{i=1}^m w_{u+1}^i$	
End	
Output: E-SMOTE-ADASVM-TW ensemble $H(x) = \text{sign}(\sum_{u=1}^U (\alpha_u f_u(x)))$	

or round down. However, the rounding functionality may also make $\sum_{j=1}^{S_P} N_u^j$ not equal to S_{N-P} , and their mantissa difference (denoted as M_u) can be calculated according to formula (7). Then the number of new synthetic samples around some original minority samples should be slightly adjusted according to formula (8) to make the total number of new synthetic samples equal to S_{N-P} . The E-SMOTE-ADASVM-TW algorithm assumes that the samples with higher weight contain more classification information for the current FDP than those with lower weight. Therefore, according to formula (8), when S_{N-P} is bigger than $\sum_{j=1}^{S_P} N_u^j$, each of the first M_u original minority class samples with higher sample weights increase one synthetic sample around it. When S_{N-P} is smaller than $\sum_{j=1}^{S_P} N_u^j$, each of the last M_u original minority class samples with lower time-based weights and with the value of N_u^j no less than one decrease one synthetic sample around it. It is worth mentioning that Z in formula (8) denotes the number of the original minority class samples whose time-based weights are the lowest and whose value of N_u^j is equal to zero. For example, if there are 10 financial distress samples ($S_P = 10$) and 50 financial normal samples ($S_N = 50$) in the initial training dataset, the total number of new synthetic financial distress samples that should be generated by SMOTE is 40 ($S_{N-P} = S_N - S_P = 50 - 10$). In the u th iteration, suppose the normalized weights of the 10 financial distress samples sorted in descending are $P_{-w_u}^j = [0.20, 0.17, 0.13, 0.13, 0.11, 0.10, 0.07, 0.05, 0.03, 0.01]$. Then the number of new synthetic samples that need to be generated around each financial distress sample should be $N_u^j = [\text{round}(40 \times 0.20), \text{round}(40 \times 0.17), \text{round}(40 \times 0.13), \text{round}(40 \times 0.13), \text{round}(40 \times 0.11), \text{round}(40 \times 0.10), \text{round}(40 \times 0.07), \text{round}(40 \times 0.05), \text{round}(40 \times 0.03), \text{round}(40 \times 0.01)] = [8, 7, 5, 5, 4, 4, 3, 2, 1, 0]$. However, at this time, $\sum_{j=1}^{S_P} N_u^j$ is equal to 39, which is less than S_{N-P} (=40), and the mantissa difference $M_u = 1$. According to formula (8), one should be added to N_u^1 , and finally $N_u^j = [9, 7, 5, 5, 4, 4, 3, 2, 1, 0]$. After $N_u^j (j = 1, 2, \dots, S_P)$ is finally determined, the SMOTE method is implemented on the original minority class dataset (denoted as D_P) to generate totally S_{N-P} synthetic minority class samples (denoted as D_SMOTE_u , which are combined with D_P to produce the training minority class dataset D_{-P_u} .

For the majority class, let $N_{-w_u}^{j'}$ represent the normalized weight of the j' th majority sample in the u th iteration of SVM ensemble training. The training majority class dataset D_{-N_u} are selected from the original majority class dataset (denoted as D_N) according to the majority samples' weights $N_{-w_u}^{j'} (j' = 1, 2, \dots, S_N)$.

Because the sample number of either D_{-P_u} or D_{-N_u} is S_N , a class-balanced training dataset D_u can be obtained by combining D_{-P_u} and D_{-N_u} . Then a base classifier SVM_u is trained based on D_u , and the base classifier's voting weight is calculated and the original samples' weights are updated with the same corresponding steps in the ADASVM-TW algorithm.

4.3. Superiority of E-SMOTE-ADASVM-TW

Hence, the E-SMOTE-ADASVM-TW model performs the SMOTE process pertinently on the basis of assigning weights to samples. For the difficult minority class samples that are misclassified by the last base classifier, SMOTE will generate more synthetic minority samples around them in the next iteration. Furthermore, the number of synthetic minority samples around newer difficult minority class samples increase by more degree than that around older difficult minority class samples. On the contrary, for the easy minority class samples that are correctly classified by the last base classifier, SMOTE will generate fewer synthetic minority samples around them in the next iteration. Furthermore, the number of synthetic minority samples around newer easy minority class samples decrease by less degree than that around older easy minority class samples. Therefore, the E-SMOTE-ADASVM-TW algorithm not only effectively balances the training dataset, but also improves the adaptation of synthetic minority class samples generated by SMOTE to the current financial distress concept in case of concept drift. At the same time, the improved sample weighting mechanism enables the misclassified and newer majority class samples to be used for base classifier training effectively, which makes the base classifiers gradually focus more on the difficult majority class samples that reflect the current FDP information. It is expected that the E-SMOTE-ADASVM-TW model can improve the performance of class-imbalanced DFDP by deep integration of SMOTE and ADASVM-TW. It makes full use of the advantage of SMOTE for treating class imbalance problem and the advantage ADASVM-TW for DFDP modeling. Meanwhile, it avoids the disadvantage of simple SMOTE that does not consider different roles of different types of samples for DFDP and the disadvantage of pure ADASVM-TW that treats the minority class samples and the majority class samples with the same steps without considering the class imbalance problem. Therefore, the E-SMOTE-ADASVM-TW model can better adapt to the class-imbalanced dynamic data flow environment, and is of great application value in the class-imbalanced DFDP.

5. Empirical experiment

We carry out empirical experiment based on the real-world data collected from Chinese listed companies, to test the effectiveness of the above approaches for class-imbalanced DFDP. This section is composed of five subsections, i.e. data collection and preprocessing, feature selection, design of empirical experiment, parameter settings, and experimental results and analysis.

5.1. Data collection and preprocessing

The empirical research is carried out based on the sample data of Chinese public companies listed in Shanghai Stock Exchange and Shenzhen Stock Exchange. The specific standard for selecting financial distress companies is that a company suffers negative net profit for two consecutive years or its net assets are less than the registered capital in the most recent fiscal year. The year in which a financial distress company is marked as distress according to its last fiscal year's financial statement is considered as the sample's standard year (T-0). The time span covered in this research is from 2002 to 2016, and some companies may have been marked as distress for several times during the period. In order to avoid repetitive sampling of the same company for several times, a company is only selected as a financial distress sample when it is marked as distress because of the above standard for the first time during the period.

To test the performance of the above DFDP approaches in the circumstance of class imbalance, the matching principle is used to select financial normal companies and the matching ratio is 1:5. In detail, the following matching criteria are used to select 5 financial normal companies for each financial distress company: Firstly, the industry is the same or similar. Secondly, the financial normal companies should have good financial performance and are not marked as distress in the three years before or after the standard year of the corresponding financial distress company. Thirdly, the book value of total assets at the end of the standard year is in the range of 30% over or below that of the corresponding financial distress company. However, if there are not sufficient financial normal companies within this range, the asset size restriction will be relaxed to select the companies satisfying the first two criteria and with the smallest difference of asset size with the corresponding financial distress company.

According to the above principles and based on the time interval from 2002 to 2016, we select 438 financial distress sample companies and 2190 financial normal sample companies that are listed in Shanghai and Shenzhen Stock Exchanges. Namely, there are totally 2628 initial sample companies, which are evidently class-imbalanced. Because this research needs to simulate the dynamic process of the sample data flow, the sample companies are divided into different batches according to their standard year, in terms of which the corresponding batch numbers are also assigned.

To construct a dataset for empirical research, 55 financial indicators were initially selected based on qualitative analysis, which considers the usage in earlier studies, the complete coverage of business operation, and the data availability. Then the sample companies' financial indicator data of the fiscal year (T-2) are collected batch by batch from China Stock Market and Accounting Research database (CSMAR).

The following preprocessing steps are implemented for a few of missing values and outliers in the initial data set: (1) A sample company's missing value of certain financial indicator is filled by this indicator's mean value of the same sample class. (2) The initial dataset may contain some outlier values that severely deviate from the mean value, i.e., data noise, which may affect the effectiveness of model training. The triple standard deviation test is adopted for outlier analysis. If a sample company's most financial indicator data deviate by more than three times of the standard deviation from the mean value, it will be deleted along with its counterpart companies. However, if a sample company only has one or several outlier values, they will also be replaced with

the mean values. Because the sample size of the data set is relatively large, a very small amount of data substitution by the mean values only has negligible effect on the empirical result.

Since different financial indicators usually have different value range, it is necessary to normalize the value range of each indicator, so as to guarantee the effectiveness of FDP modeling. In this study, the financial indicators are normalized according to formula (9), so that the range of values of each financial indicator is maintained between $[-1, 1]$. Herein, $\min_V$ and $\max_V$ are respectively the minimum and maximum values of certain financial indicator.

$$V' = -1 + \frac{V - \min_V}{\max_V - \min_V} \times [1 - (-1)] \quad (9)$$

5.2. Feature selection

Feature selection includes the following three steps. Firstly, initial features are selected. Secondly, normality distribution is tested and the mean values of financial distress samples and financial normal samples are statistically compared for each feature. Thirdly, the final features for DFDP modeling is selected by stepwise discriminant analysis and collinearity test.

5.2.1. Initial features

The initial features include 55 financial indicators. They cover the eight aspects of profitability, operating capacity, solvency, structural ratios, growth ability, risk level, per share ratios, and shareholder profitability, as shown in Table 3.

5.2.2. Normality test and mean comparison

Since the sample size is much greater than 50, the Kolmogorov–Smirnov test is used to determine whether the financial indicators obey the normal distribution. The test results show that none of the alternative financial indicators obey normal distribution. Therefore, the Mann–Whitney non-parametric test method was used for mean comparison, so as to select the financial indicators that have significantly different means between the financial distress samples and the financial normal samples. It is shown only two financial indicators, i.e. current liabilities to total liabilities ratio (V34) and price to net assets ratio (V54), are not significant at the significance level of 5%, and they are deleted from the alternative features.

5.2.3. Stepwise discriminant analysis and collinearity test

The remaining financial indicators are then filtered by stepwise discriminant analysis, and those with significant discriminant ability for the two classes are selected out, as shown in the first column of Table 4. Since the financial indicators are figured out from the items in the financial statements, there may be certain degree of correlation between two financial indicators. In order to ensure the rationality of FDP model, it is necessary to measure the correlation between the financial indicators and avoid multi-collinearity among them. In this paper, multi-collinearity is tested by the statistics of tolerance (TOL) and variance inflation factor (VIF), and the results are shown in Table 4. It is generally believed that there is multi-collinearity between variables when TOL is less than 0.1 or VIF is greater than 10. As shown in Table 4, there is no multi-collinearity for the 19 financial indicators selected by stepwise discriminant analysis, which constitute the input variables for FDP modeling.

5.3. Descriptive statistics

The descriptive statistics of the 19 financial indicators finally selected for FDP modeling is listed in Table 5, in which FD mean and NM mean represent the mean value of financial distress samples and financial normal samples. Obviously, the value range of different features before normalization differ greatly, and particularly V13, V37 and V52 have much larger value range than the other features, which also proves

Table 3
Initial financial indicators.

Types	Variables and financial indicators
Profitability	V1: Gross profit to sales ratio, V2: Net profit to operating revenue ratio, V3: EBIT to total assets ratio, V4: Net profit to total assets ratio, V5: Net profit to current assets ratio, V6: Net profit to fixed assets ratio, V7: Net profit to owners' equity ratio, V8: return on invested capital, V9: EBIT to long-term capital, V10: Return on cost and expense
Operating capacity	V11: Accounts receivable turnover, V12: Inventory turnover, V13: Operating cycle, V14: Accounts payable turnover, V15: Current assets turnover, V16: Fixed assets turnover, V17: Total assets turnover, V18: Capital intensity ratio
Solvency	V19: Current ratio, V20: Quick ratio, V21: Conservative quick ratio, V22: Cash flow ratio, V23: Interest coverage ratio, V24: Asset-liability ratio, V25: Owners' equity to liabilities ratio, V26: Profit before interest, depreciation and amortization to liabilities ratio, V27: Net cash flow from operating activities to liabilities ratio, V28: Liabilities to market value of owners' equity ratio, V29: Liabilities to tangible net worth ratio
Structural ratios	V30: Current assets to total assets ratio, V31: Fixed assets to total assets ratio, V32: Long-term capital to fixed assets and long-term investment ratio, V33: Owners' equity ratio to fixed assets, V34: Current liabilities to total liabilities ratio
Growth ability	V35: Fixed assets, V36: Total assets growth rate, V37: Net profit growth rate, V38: Operating revenue growth rate
Risk level	V39: Financial leverage coefficient, V40: Operating leverage coefficient, V41: Total leverage coefficient
Per share ratios	V42: Earnings per share, V43: Operating revenue per share, V44: Net assets per share, V45: Tangible assets per share, V46: Surplus reserves per share, V47: Undistributed profit per share, V48: Net cash flows from operating activities per share, V49: Net cash flows from investing activities per share, V50: Net cash flows from financing activities per share, V51: Net cash flows per share
Shareholder profitability	V52: Price to earnings ratio, V53: Price to cash flow ratio, V54: Price to net assets ratio, V55: Price to tangible assets ratio

Table 4
Results of multi-collinearity test.

Financial indicators	Statistics for multi-collinearity test	
	TOL	VIF
V2: Net profit to operating revenue ratio	0.312	3.210
V3: EBIT to total assets ratio	0.166	6.022
V5: Net profit to current assets ratio	0.204	4.892
V6: Net profit to fixed assets ratio	0.598	1.672
V7: Net profit to owners' equity ratio	0.186	5.371
V9: EBIT to long-term capital	0.198	5.049
V10: Return on cost and expense	0.292	3.423
V13: Operating cycle	0.595	1.680
V18: Capital intensity ratio	0.656	1.524
V28: Liabilities to market value of owners' equity ratio	0.487	2.052
V29: Liabilities to tangible net worth ratio	0.651	1.536
V31: Fixed assets to total assets ratio	0.802	1.247
V37: Net profit growth rate	0.768	1.302
V40: Operating leverage coefficient	0.943	1.061
V41: Total leverage coefficient	0.855	1.170
V42: Earnings per share	0.237	4.217
V47: Unallocated profit per share	0.630	1.588
V52: Price to earnings ratio	0.887	1.127
V55: Price to tangible assets ratio	0.568	1.762

Table 5
Descriptive statistics for the selected financial indicators.

Financial indicators	Before normalization				After normalization			
	Min	Max	FD mean	NM mean	Min	Max	FD mean	NM mean
V2	−2.97	1.20	−0.22	0.08	−1	1	0.32	0.46
V3	−0.35	0.21	−0.05	0.05	−1	1	0.08	0.43
V5	−0.86	0.50	−0.14	0.08	−1	1	0.05	0.38
V6:	−11.35	4.96	−0.34	0.24	−1	1	0.35	0.42
V7	−1.36	0.38	−0.16	0.07	−1	1	0.38	0.65
V9	−0.99	0.47	−0.09	0.10	−1	1	0.24	0.50
V10	−1.00	0.77	−0.15	0.12	−1	1	−0.04	0.26
V13	0.00	4761.79	493.12	236.01	−1	1	−0.79	−0.90
V18	0.11	28.67	3.75	2.47	−1	1	−0.74	−0.83
V28	0.00	3.16	0.70	0.40	−1	1	−0.56	−0.75
V29	0.01	18.65	2.22	1.08	−1	1	−0.76	−0.88
V31	0.00	0.79	0.31	0.27	−1	1	−0.23	−0.31
V37	−275.93	165.69	−12.69	−0.12	−1	1	0.19	0.25
V40	−42.04	35.79	0.62	1.54	−1	1	0.10	0.12
V41	−33.28	59.00	0.66	2.49	−1	1	−0.26	−0.22
V42	−1.86	1.27	−0.35	0.24	−1	1	−0.04	0.34
V47	−1.98	3.06	−0.09	0.54	−1	1	−0.25	0.00
V52	−5082.64	10,041.89	−7.51	87.81	−1	1	−0.33	−0.32
V55	0.10	7.27	1.50	2.01	−1	1	−0.61	−0.47

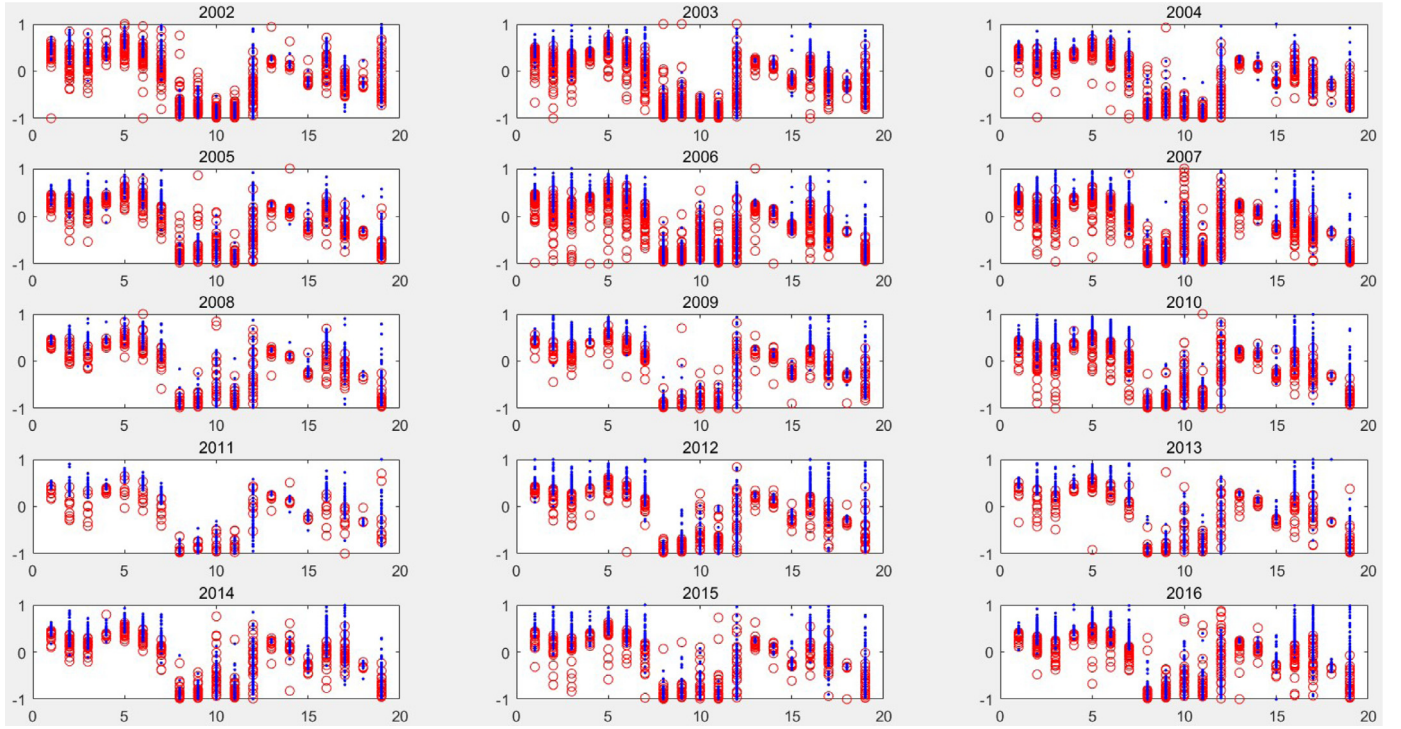


Fig. 4. The sample graph of the time series.

the necessity of normalization. Besides, the FD mean and the NM mean show significant difference, and the financial indicators related to profit show negative mean values for financial distress samples and positive mean values for financial normal samples before normalization. To further illustrate the variation of sample distribution with time going on, we graph the sample distribution of each feature after normalization year by year and show it as Fig. 4, in which the lateral axis means different features and the vertical axis means feature value, and the red circles represent financial distress samples and the blue dots represent financial normal samples. It indicates that the data distribution changes gradually year by year, which also proves that the virtual concept drift exists for the problem of FDP.

5.4. Design of empirical experiment

To test the effectiveness of the proposed approach, we design the experimental process to simulate the DFDP in the real world, and introduce the model evaluation measures that are suitable for class-imbalanced DFDP in this section.

5.4.1. Design of experimental process

In order to deal with concept drift, it is necessary to simulate the dynamic flow of the data stream over time. The experiment uses one year as the basic time unit, and the current year is supposed to transit from 2005 to 2015. When time moves forward each year, the new samples of financial distress companies and the corresponding financial normal companies should be collected and their financial indicator data constitute a new data batch, which is added to the data stream so as to simulate the dynamic data flow in the real world. In the experiment, the FDP models are built based on the T-2 data of the sample companies available at the current modeling year, meanwhile, the models are tested with the T-2 data of the sample companies that emerge one year after the modeling year. For example, when the current year is 2005, the T-2 data of the sample companies collected from 2002 to 2005 are used as training set 1, and the T-2 data of sample companies collected in 2006 are used as testing set 1, so as to evaluate the prediction performance of the FDP model constructed in 2005. Similarly, when the time

advances to 2006, the T-2 data of sample companies collected in 2006 are added to the training set so that the T-2 data of sample companies collected from 2002 to 2006 are used for model training. Then, the T-2 data of sample companies collected in 2007 are used as testing set 2. By this way, as time passes, the data flow is constantly updated, and DFDP is simulated with time moving on, as shown in Fig. 5.

The DFDP modeling and testing experiments were implemented using Matlab2015b software. Since there is certain degree of randomness in the proposed model, 50 times of experiments were performed for each model under the same computational environment, in order to obtain more objective and comprehensive experimental results for more scientific comparison and analysis of model performance.

5.4.2. Model evaluation measure

In most FDP studies based on class-balanced data sets, the ratio of correctly classified sample number to the total sample number, i.e. the overall accuracy rate (abbreviated as *Acc*), is widely used as the model evaluation measure. However, when the samples are class imbalanced, this measure cannot reflect a model's recognition performance for the minority class well, and it is even far from the real classification accuracy of financial distress samples when the two classes are highly class-imbalanced. For example, there are 10 financial distress samples in 100 total samples, and the rest are financial normal samples. If all the 10 financial distress samples are misclassified and 90 financial normal samples are all classified correctly, *Acc* is 90% but the accuracy rate for financial distress samples is zero. Obviously, for the class-imbalanced samples, *Acc* is not suitable to be the evaluation measure alone. Therefore, this study also adopts some other commonly used evaluation measures for class-imbalanced classification models, including minority class accuracy rate (also called as true positive rate or recall, abbreviated as *TPrate*), majority class accuracy rate (also called as true negative rate, abbreviated as *TNrate*), *F* value and *G* value, which can be calculated according to formula (10) to formula (14). In the formulas, the meanings of *TP*, *TN*, *FP* and *FN* are shown in Table 6.

$$TPrate(recall) = TP / (TP + FN) \quad (10)$$

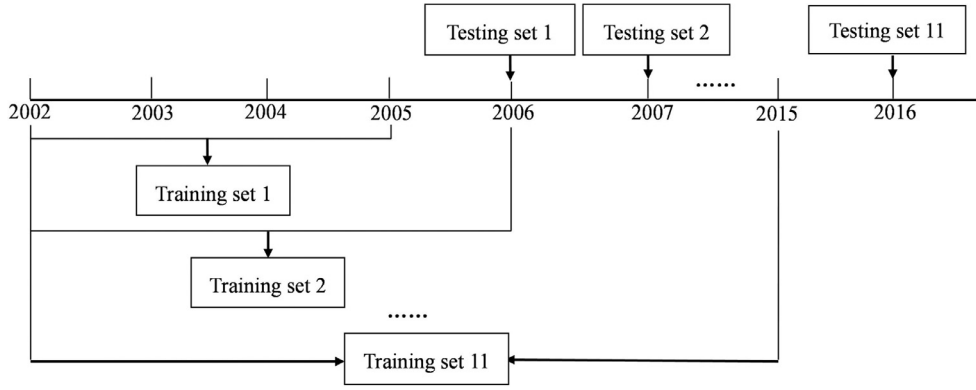


Fig. 5. The training sets and testing sets for simulated DFDP with time moving on.

Table 6
The confusion matrix.

Actual classes	Prediction results	
	Positive class (Financial distress)	Negative class (Financial normal)
Positive class (Financial distress)	TP	FN
Negative class (Financial normal)	FP	TN

$$TNrate = TN / (TN + FP) \quad (11)$$

$$precision = TP / (TP + FP) \quad (12)$$

$$F = \frac{(1 + \beta^2) \times recall \times precision}{\beta^2 \times recall + precision} \quad (13)$$

$$G = \sqrt{\frac{TP}{TP + FN} \times \frac{TN}{TN + FP}} \quad (14)$$

In the above formulas, *recall* and *precision* represent the recall and precision of the minority class, and the parameter β usually takes the value of 1. The above formulas show that only when the recall and precision are high at the same time can *F* value achieves a high rate, indicating that *F* value can reflect the classification performance for the minority class. Similarly, it is only when the *TPrate* and *TNrate* are large at the same time, will *G* value achieves a larger value. Namely, when *G* value is large, both classes can be recognized with relatively high accuracy. Therefore, *F* value and *G* value both consider the prediction performance of the minority class, and they are suitable for measuring the performance of a class-imbalanced FDP model. In addition, *TPrate* or *TNrate* respectively represents the percentage of correctly classified samples inside each class, and can also intuitively reflect the model's discriminative performance for each class on the imbalanced dataset [48].

5.5. Parameter settings

There are two parameters, λ and c , in the algorithms of ADASVM-TW, S-SMOTE-ADASVM-TW, and E-SMOTE-ADASVM-TW. They should be determined before the formal experiment is executed.

5.5.1. Determination of the parameter λ

The parameter λ is an important parameter of the time weighting model, whose value affects the model's effectiveness, and it is necessary to find an optimal parameter value before the formal experiment. According to the findings of Sun et al. [14], the search range of the parameter λ is set as [0.18, 0.72]. A pre-experiment is carried out for determination of λ before the formal experiment starts. In the formal experiment, the current modeling year is supposed to evolve from 2005 to 2015, and

the testing year correspondingly evolves from 2006 to 2016. While in the pre-experiment, we suppose 2004 as the current modeling year and 2005 as the testing year. Namely, the T-2 data of 2002–2004 are used as the training set, and the T-2 data of 2005 are used as the testing set in the pre-experiment. Then the testing performance under different λ values in the pre-experiment is compared, and the parameter value corresponding to the best performance is selected as the optimal parameter. Taking the ADASVM-TW model as an example, for $\lambda \in \{0.18, 0.21, \dots, 0.72\}$, we illustrate the testing results of grid search with the step of 0.03 in Fig. 6. It shows that when λ takes the value of 0.33, the testing performance is overall best, and the parameter λ for the ADASVM-TW model is set as 0.33 in the formal experiment. Through similar process, the optimal values of λ for the S-SMOTE-ADASVM-TW model and the E-SMOTE-ADASVM-TW model are set as 0.57 and 0.42 respectively.

5.5.2. Determination of the threshold c

The threshold c in the algorithms of ADASVM-TW, S-SMOTE-ADASVM-TW, and E-SMOTE-ADASVM-TW is used to control the error rate of the base SVM classifier on the training data set, to avoid over-fitting. Since the training data set for base classifier training is already class-balanced by the processing of SMOTE, the error rate directly equals to one minus the accuracy rate. Ni et al. [56] points out that the FDP accuracy of Chinese listed companies based on the SVM model is between 68% and 92%, which is basically consistent with the results of other researches. In the pre-experiment of SVM training based on the dataset of this study, the highest accuracy rate is about 96%. Therefore, the error rate threshold c is set as 4%.

5.6. Experimental results and analysis

With the supposed current modeling year transiting from 2005 to 2015, the corresponding prediction year transits from 2006 to 2016, and the testing results of overall accuracy rate, *TPrate*, *TNrate*, *F* value and *G* value are listed in Table 7, in which each number is calculated as the average of fifty times of experiments and the mean in the last column is the average of each measure for all the prediction years weighted by the testing sample number of the corresponding year. For each performance measure, the numbers in bold are the maximum values among the three models and the numbers underlined are the minimum values among the three models. For better illustration, the mean *Acc*, *TPrate*, *TNrate*, *F* value and *G* value in the last column of Table 7 are intuitively

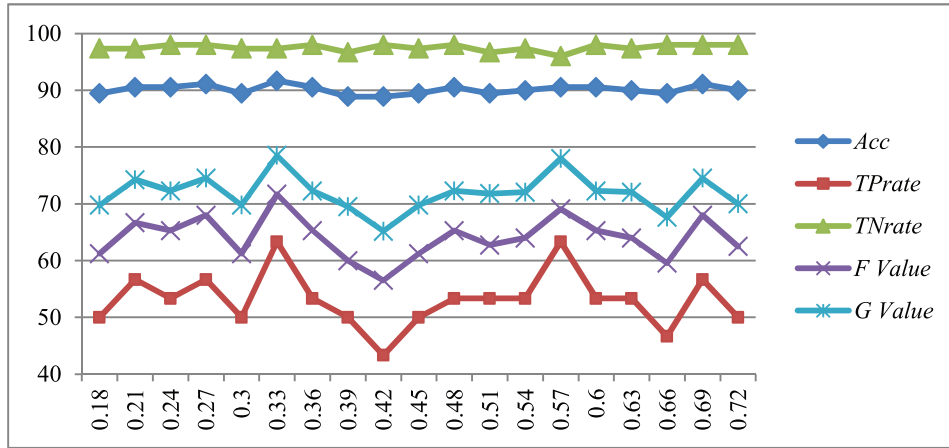


Fig. 6. The searching results of parameter λ for the ADASVM-TW model.

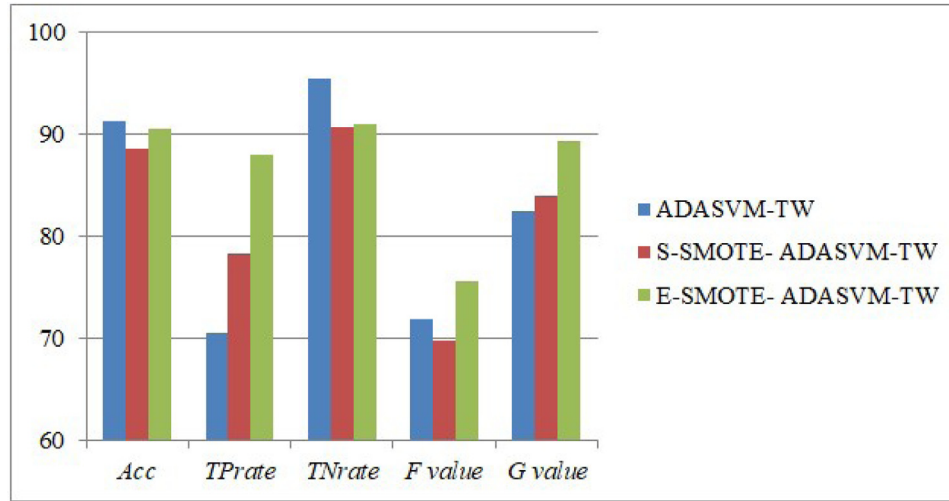


Fig. 7. The column graph of the mean values of performance measures.

Table 7
The testing results of *Acc*, *TPrate*, *TNrate*, *F value* and *G value* (%).

Models		Prediction years											
		2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	Mean
<i>Acc</i>	ADASVM-TW	92.65	94.08	90.54	<u>90.33</u>	88.57	93.81	<u>90.08</u>	89.77	85.82	90.71	93.94	91.22
	S-SMOTE- ADASVM-TW	86.23	90.67	88.50	91.65	<u>85.12</u>	89.04	91.60	88.58	85.26	90.78	90.33	88.63
	E-SMOTE- ADASVM-TW	88.97	90.94	88.80	91.62	92.23	90.93	92.05	91.79	85.96	91.63	91.79	90.47
<i>TPrate</i>	ADASVM-TW	<u>70.20</u>	84.17	<u>50.33</u>	46.30	<u>64.06</u>	76.00	<u>54.67</u>	<u>75.00</u>	78.23	<u>73.62</u>	<u>82.97</u>	<u>70.54</u>
	S-SMOTE- ADASVM-TW	80.80	89.88	60.22	60.40	85.74	68.44	67.14	78.00	74.08	78.10	85.24	78.26
	E-SMOTE- ADASVM-TW	88.12	91.63	66.78	80.00	93.16	77.78	84.76	92.38	91.77	84.29	97.72	88.03
<i>TNrate</i>	ADASVM-TW	97.14	96.07	98.58	99.14	93.47	97.38	97.16	92.73	87.34	94.13	96.14	95.36
	S-SMOTE- ADASVM-TW	<u>87.32</u>	90.83	94.16	97.90	<u>84.99</u>	<u>93.16</u>	96.50	<u>90.70</u>	87.49	93.33	91.35	<u>90.70</u>
	E-SMOTE- ADASVM-TW	89.14	90.81	93.20	93.94	92.04	93.56	93.50	91.68	84.80	93.10	90.61	90.95
<i>F value</i>	ADASVM-TW	76.08	82.57	63.91	61.24	65.05	80.45	64.67	71.01	64.77	<u>72.51</u>	82.04	71.91
	S-SMOTE- ADASVM-TW	66.24	76.31	63.61	70.52	65.89	67.71	72.62	69.47	62.62	73.90	<u>74.75</u>	<u>69.78</u>
	E-SMOTE- ADASVM-TW	72.77	77.17	66.55	76.11	79.98	74.10	78.03	80.61	68.55	77.02	80.13	75.64
<i>G value</i>	ADASVM-TW	<u>82.50</u>	<u>89.90</u>	<u>70.40</u>	<u>67.59</u>	<u>77.30</u>	85.99	<u>72.79</u>	<u>83.29</u>	82.62	<u>83.20</u>	89.27	<u>82.40</u>
	S-SMOTE- ADASVM-TW	83.98	90.34	75.27	76.76	85.35	79.81	80.42	84.01	80.49	85.34	88.22	83.96
	E-SMOTE- ADASVM-TW	88.62	91.21	78.89	86.69	92.60	85.30	89.00	91.64	88.21	88.55	94.08	89.33
Testing sample number		300	288	108	120	186	54	126	96	156	126	174	–

illustrated with column graph in Fig. 7 and their corresponding values obtained respectively in fifty times of experiments are graphed as curves in Fig. 8.

Shapiro-Wilk test is carried out for each model's performance measures obtained in the 50 times of experiments, and the results show that most of them follow normal distribution except *TPrate* of S-SMOTE-ADASVM-TW and *Acc*, *TNrate* and *G value* of E-SMOTE-ADASVM-TW (see Table 8 for detail). To further test whether there is significant dif-

ference among the three models' performance, both the paired-sample *T*-test and the nonparametric Wilcoxon test were carried out based on the values of performance measures obtained in the 50 times of experiments, and the results are listed in Table 9. It is shown that the results of the paired-sample *T*-test and the nonparametric Wilcoxon test are almost the same except the comparison of *TNrate* between S-SMOTE-ADASVM-TW and E-SMOTE-ADASVM-TW, and the mean values of performance measures between each pair of the above models are significantly

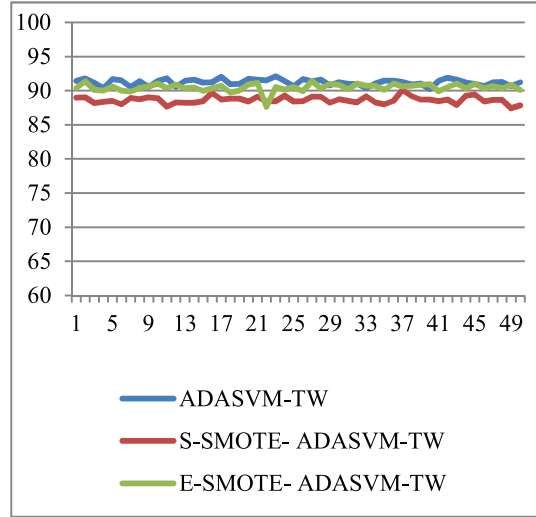
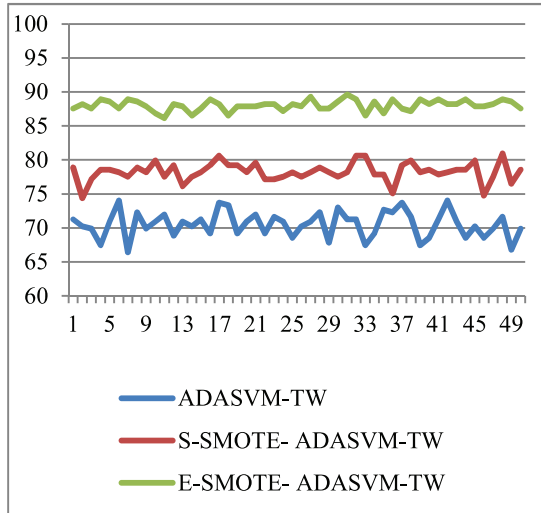
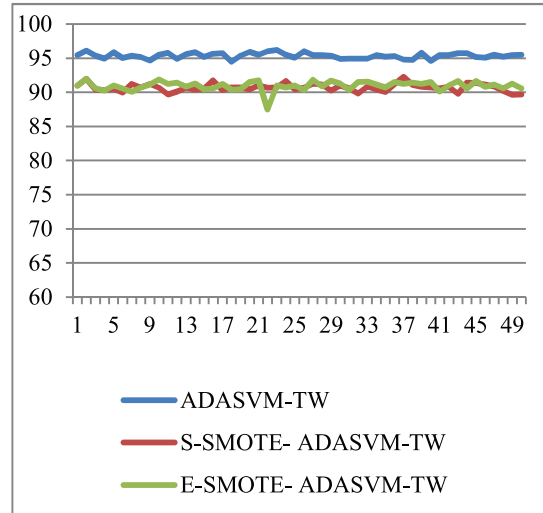
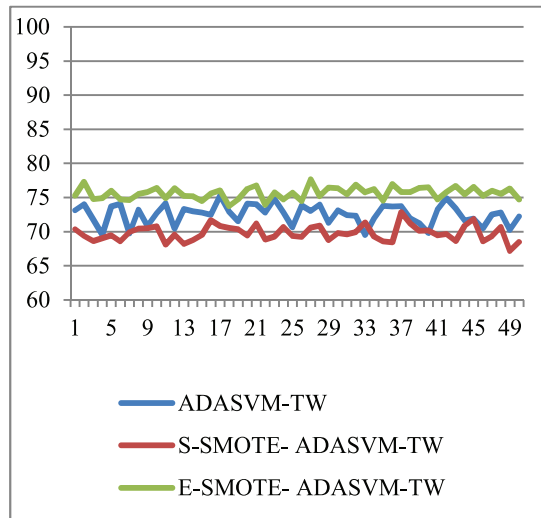
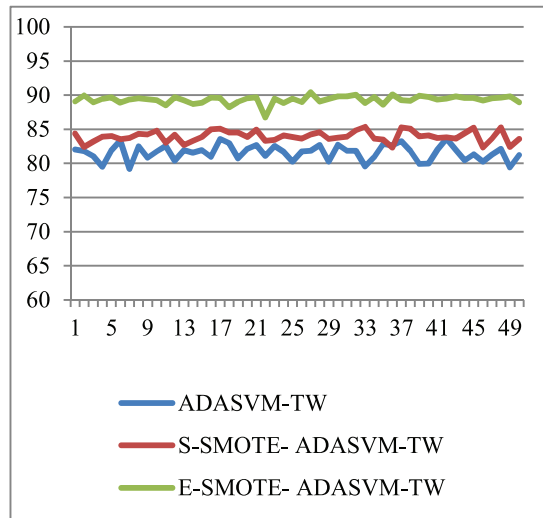
(a) Mean *Acc* in fifty times of experiments(b) Mean *TPRate* in fifty times of experiments(c) Mean *TNrate* in fifty times of experiments(d) Mean *F* in fifty times of experiments(e) Mean *G* in fifty times of experiments**Fig. 8.** Curves of mean *Acc*, *TPRate*, *TNrate*, *F* value and *G* value in fifty times of experiments.

Table 8
Significance values of Shapiro–Wilk test for model performance measures.

Model	Acc	TPrate	TNrate	F value	G value
ADASVM-TW	0.396	0.454	0.814	0.101	0.249
S-SMOTE-ADASVM-TW	0.833	0.024	0.594	0.736	0.163
E-SMOTE-ADASVM-TW	0.000	0.075	0.000	0.822	0.000

different at the level of 1% except the *TNrate* between the S-SMOTE-ADASVM-TW model and the E-SMOTE-ADASVM-TW model.

In terms of the *Acc* results, the ADASVM-TW model has higher mean *Acc* than the S-SMOTE-ADASVM-TW model and the E-SMOTE-ADASVM-TW model respectively at the level of 1%, and the E-SMOTE-ADASVM-TW model has higher mean *Acc* than the S-SMOTE-ADASVM-TW model at the level of 1%. In detail, the ADASVM-TW model obtains the highest mean *Acc* value, and it has the highest overall accuracies for five prediction years and the lowest overall accuracies for two prediction years. The S-SMOTE-ADASVM-TW model shows the lowest overall accuracies in most years and its mean *Acc* value is also the lowest among the three models. The E-SMOTE-ADASVM-TW model has the intermediate mean *Acc* value, and meanwhile it achieves the highest overall accuracies for five prediction years without the lowest overall accuracies for all the prediction years, indicating that the E-SMOTE-ADASVM-TW model has the most stable *Acc*.

According to the *TPrate* results, the E-SMOTE-ADASVM-TW model has higher mean *TPrate* than the S-SMOTE-ADASVM-TW model, and the S-SMOTE-ADASVM-TW model has higher mean *TPrate* than the ADASVM-TW model, respectively at the level of 1%. In detail, the ADASVM-TW model has the lowest mean accuracy for financial distress samples, and it has the lowest *TPrate* values for nine prediction years without any highest *TPrate* values. Such results indicate that the ADASVM-TW model without class imbalance treatment is not suitable for class-imbalanced DFPD because of low recognition capability for the minority financial distress samples. The mean *TPrate* values of the S-SMOTE-ADASVM-TW model and the E-SMOTE-ADASVM-TW model are both much higher than that of the ADASVM-TW model, which means that either simple integration of SMOTE or embedding integration of SMOTE with the ADASVM-TW model can greatly improve the accuracy for the minority financial distress samples. Furthermore, the E-SMOTE-ADASVM-TW model achieves the highest mean *TPrate* value among the three models, as well as the highest *TPrate* values for all the prediction years.

As for the *TNrate* results, the ADASVM-TW model has higher mean *TNrate* than the S-SMOTE-ADASVM-TW model and the E-SMOTE-ADASVM-TW model at the level of 1%, and the mean *TNrate* values of the latter two models are not significantly different. In detail, the ADASVM-TW model has the highest mean accuracy for the majority financial normal samples, and it has the highest *TNrate* values for ten prediction years, which is also the reason for the ADASVM-TW model's high overall accuracies in spite of very low *TPrate* values. The possible reason

is that the ADASVM-TW model without class imbalance treatment shows bias on the majority class in case of high degree of class imbalance, thereby reducing the classification accuracy for the minority class. The S-SMOTE-ADASVM-TW model and the E-SMOTE-ADASVM-TW model have the similar mean *TNrate* values, which are lower than that of the ADASVM-TW model. It is clear that either the S-SMOTE-ADASVM-TW model or the E-SMOTE-ADASVM-TW model sacrifices some degree of recognition capability for the majority financial normal samples to improve the recognition capability for the minority financial distress samples.

In terms of *F* value, the E-SMOTE-ADASVM-TW model has higher mean *F* value than the ADASVM-TW model, and the ADASVM-TW model has higher mean *F* value than the S-SMOTE-ADASVM-TW model, respectively at the level of 1%. In detail, the E-SMOTE-ADASVM-TW model achieves the highest mean *F* value, and it shows the highest *F* values for eight prediction years and the intermediate *F* values for the remaining three prediction years. However, the S-SMOTE-ADASVM-TW model suffers the lowest mean *F* value, and it shows the lowest *F* values for seven prediction years and the intermediate *F* values for the remaining four prediction years. The ADASVM-TW model without class imbalance treatment gets the intermediate mean *F* value, which is a little higher than that of the S-SMOTE-ADASVM-TW model and much lower than that of the E-SMOTE-ADASVM-TW model. Since the *F* value is the aggregation measure based on *TPrate* and *precision*, it can be inferred that the improvement of *TPrate* by the S-SMOTE-ADASVM-TW model is not sufficient enough to balance certain degree of sacrifice of *precision*. However, the E-SMOTE-ADASVM-TW model can improve *TPrate* with large degree, which overwhelms the degree of sacrifice of *precision*.

For *G* value, the E-SMOTE-ADASVM-TW model has higher mean *G* value than the S-SMOTE-ADASVM-TW model, and the S-SMOTE-ADASVM-TW model has higher mean *G* value than the ADASVM-TW model, respectively at the level of 1%. In detail, the E-SMOTE-ADASVM-TW model achieves the highest mean *G* value, and it shows the highest *G* values for ten prediction years and the intermediate *G* values for the remaining one prediction year. The ADASVM-TW model has the lowest mean *G* value, and it has the lowest *G* values for eight prediction years. The S-SMOTE-ADASVM-TW model has the intermediate mean *G* value, which is only a little higher than that of the ADASVM-TW model. Therefore, the results of *G* value almost accord with those of *F* value, and they both support that the E-SMOTE-ADASVM-TW model has better overall performance and is more suitable for class-imbalanced DFPD than the S-SMOTE-ADASVM-TW model.

In summary, compared with the ADASVM-TW model, the S-SMOTE-ADASVM-TW model significantly improves *TPrate* and *G* value at the expense of significantly sacrificing *Acc*, *TNrate* and *F* value, and the E-SMOTE-ADASVM-TW model significantly improves *TPrate*, *F* value and *G* value at the expense of significantly sacrificing *Acc* and *TNrate*. The possible reason is that the FDP models built by ADASVM-TW without class imbalance treatment severely bias towards the majority non-financial distress samples, and either S-SMOTE-ADASVM-TW or

Table 9
The statistical significance results for the mean comparison of performance measures.

Modell names	Measures	S-SMOTE-ADASVM-TW		E-SMOTE-ADASVM-TW	
		T-test	Wilcoxon test	T-test	Wilcoxon test
ADASVM-TW	Acc	0.000	0.000	0.000	0.000
	TPrate	0.000	0.000	0.000	0.000
	TNrate	0.000	0.000	0.000	0.000
	F value	0.000	0.000	0.000	0.000
	G value	0.000	0.000	0.000	0.000
S-SMOTE-ADASVM-TW	Acc	–	–	0.000	0.000
	TPrate	–	–	0.000	0.000
	TNrate	–	–	0.052	0.020
	F value	–	–	0.000	0.000
	G value	–	–	0.000	0.000

E-SMOTE-ADASVM-TW can mitigate such model bias to some extent by generating new synthetic financial distress samples. Furthermore, E-SMOTE-ADASVM-TW generates the new synthetic financial distress samples with a sample weighting mechanism, which are more biased towards newer and more difficult samples. However, S-SMOTE-ADASVM-TW generates the same number of new synthetic financial distress samples around each original financial distress sample, which may generate more noisy samples when concept drifts. Therefore, E-SMOTE-ADASVM-TW outperforms S-SMOTE-ADASVM-TW for the class-imbalanced DFDP with financial distress concept drift.

6. Conclusion

How to effectively construct the FDP model is always an important research topic for corporate financial management. Considering the effect of financial distress concept drift, some DFDP modeling approaches have been proposed to dynamically construct FDP models based on the financial data streams. However, they almost ignore the class imbalance problem in the process of DFDP modeling. This paper tries to consider this problem and propose two class-imbalanced DFDP models based on Adaboost-SVM ensemble combined with SMOTE and time weighting, i.e. the S-SMOTE-ADASVM-TW model and the E-SMOTE-ADASVM-TW model. The S-SMOTE-ADASVM-TW model simply applies SMOTE to balance all the available data batches before the ADASVM-TW algorithm is used to construct the Adaboost-SVM ensemble model for DFDP. The E-SMOTE-ADASVM-TW model embeds SMOTE into the iteration of the ADASVM-TW model, treating the new and old minority samples differently as well as treating the difficult and easy minority samples differently in the iteration of base classifier training. The empirical experiment is carried out based on the sample data of totally 2628 Chinese listed companies, which consist of 438 financial distress samples and 2190 financial normal samples. Experimental results show that the ADASVM-TW model without SMOTE has very bad performance in recognizing the minority class of financial distress samples, and both S-SMOTE-ADASVM-TW and E-SMOTE-ADASVM-TW can solve this problem to some extent. Furthermore, the E-SMOTE-ADASVM-TW model significantly outperforms the S-SMOTE-ADASVM-TW model and is more preferred for class-imbalanced DFDP.

In the future research, these two models can be further applied to solving the other problems such as default diagnosis, customer classification, spam filtering, etc., which may also show the characteristics of class imbalance and concept drift. Besides, the dynamic sample weighting mechanism suitable for the integration of SMOTE with other ensemble learning algorithms, e.g. Random Space and Bagging, can be further explored, so as to propose more effective classifier ensemble approaches for class-imbalanced non-stationary classification.

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